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MTH621Real Analysis IMid-Term PreparationImportant Theorems For shortQuestions & For ObjectiveChapter #01The Real Number System $\Rightarrow$ proper subset; A is proper subset of B ifthere exist at least one element of B which is not in A . $\Rightarrow$ Equal sets;

"Two sets are said to be equal if they contain the same number of elements."

 $\Rightarrow$ Set of odd Number;

$$\{2k-1 : k \in \mathbb{N}\}$$

 $\Rightarrow$ Set of Even number;

$$\{2k : k \in \mathbb{N}\}$$

 $\Rightarrow$ A set with only one member x_0 is Singleton set, denoted by $\{x_0\}$.

$\Rightarrow$ Prime numbers are the building blocks of the positive integers.

$\Rightarrow$ An equation with the added provision that only integer solutions are sought is called Diophantine.

$$\text{i.e. } a^n + b^n = c^n, n \in \mathbb{Z}$$

$\Rightarrow$ Mathematical Induction;

let $P(n)$ be a mathematical statement, where $n \in \mathbb{N}$ (or $\mathbb{Z}^+$) i.e.

① $P(1)$ is true

② $P(n)$ is true implies $P(n+1)$ is true.

Examples;

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$1^3 + 2^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$$

$\Rightarrow$ An initial segment:

An initial segment I of $\mathbb{N}$ is a non-empty subset of $\mathbb{N}$ with the property that if $n \in I$ and $m \leq n$ then $m \in I$.

② Finite and Infinite set;

A set A is Finite if either A is empty or there exists $n \in \mathbb{N}$ and a Bijective mapping $c: I_n \rightarrow A$.

OR

A set S is said to be Finite if it is either empty or it has n elements for some $n \in \mathbb{N}$.

$\Rightarrow$ Infinite :-

A set is $\neq$ infinite if it is not Finite.

$\Rightarrow$ If $g: I_m \rightarrow I_n$ is an one to one (Injective) mapping then $m \leq n$.

$\Rightarrow$ If A is a non-empty finite set, there exist a unique $n \in \mathbb{N}$ for which there exist a bijection $c: I_n \rightarrow A$.

$\Rightarrow$ The Number n is known as Cardinality of A and is written as $|A|$.

$\Rightarrow$ Proposition :-

Suppose (A) is Finite set, and that $f: A \rightarrow B$ is a bijective. Then B is Finite, and $|B| = |A|$

Proof:- For if $C: I_{|A|}^{-1}A$ is a bijective, then the mapping

$f \circ C: I_{|A|}^{-1}B$ is the bijection.

$\Rightarrow$ Field;

A Field is a set F , together with two laws of composition, addition (+) and Multiplication ($\cdot$) such that for all $a, b, c \in F$ following properties holds;

- * Commutative * associative * distributive
- * Identity. * Inverse.

$\Rightarrow$ Dedekind Infinite set;

A set A to be infinite if there is an injective map $j: A \rightarrow A$ which is not onto (surjective), such sets are called Dedekind Infinite.

$\Rightarrow$ Example:- Show that N is Dedekind infinite

Sol; The mapping $f: N \Rightarrow N$ defined by $f(n) = 2n$ is injective, and not surjective.

③ $\Rightarrow$ The set of Natural numbers N is infinite set.

$\Rightarrow$ Countable, If it is finite set. otherwise uncountable.

$\Rightarrow$ Not every set is countable, It was Cantor who first showed in 1873.

$\Rightarrow$ If A be a non-empty set. Then the following are equivalent;

(a) A is countable.

(b) There exists a surjection $f: N \rightarrow A$.

(c) There exists an injection $g: A \rightarrow N$.

$\Rightarrow$ Theorem; The set $N \times N$ is countable.

Proof; The mapping $f: N \times N \rightarrow N$ by

$$f(k, l) = 2^k 3^l.$$

$N \times N$ is countable.

$\Rightarrow$ The set of Rational Numbers is defined

as, $Q = \left\{ \frac{m}{n} ; m, n \in \mathbb{Z}, n \neq 0 \right\}$

Example;

Find solution of the equation

$p = \sqrt{2}$ in the set of Rational Numbers
if possible;

Proof:-

IF $\sqrt{2}$ is Rational Number.

$$\text{So, } \sqrt{2} = \frac{p}{q}$$

Taking square on both sides;

$$2 = \frac{p^2}{q^2} \Rightarrow p^2 = 2q^2 \rightarrow \textcircled{1}$$

p^2 should be of the form $4p'^2$

$$4p'^2 = 2q^2$$

$$2p'^2 = q^2$$

$$2p'^2 = 4q'^2$$

$$p^2 = 2q'^2$$

$$p = 2p'$$

$$q = 2q'$$

$$\frac{p}{q} = \frac{2p'}{2q'}$$

Hence $\sqrt{2}$ is an irrational Number.

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⇒ Partial Order;

A Partial order is a relation R on a set S such that for all $a, b, c \in S$ we have the following;

* Reflexivity (aRa)

* Anti symmetric (if aRb and $bRa \Rightarrow a=b$)

* Transitive ($aRb, bRc \Rightarrow aRc$)

⇒ Lower Bound;

Any smaller number of set.
if b is a lower bound of E , then
so is any smaller number.

⇒ Infimum:-

if α is a lower bound of E ,
but no number greater than α . Then α
is an infimum of E , and we write.

$$\alpha = \inf E$$

⇒ Supremum; upper Bound:-

if b is an upper bound of E ,
then so, is any larger number.

⇒ Supremum;

if B is an upper bound of E
but no number is less than B is an upper
Bound of E , then B is supremum of E .

$$B = \sup E.$$

$\Rightarrow$ If S is the set of negative numbers, then any nonnegative number is an upper bound of S and $\sup S = 0$.

$\Rightarrow$ If S is the set of negative integers, then any number a such that $a \geq -1$ is an upper bound of S , and $\sup S = -1$.

$\Rightarrow$ The supremum and infimum of a set may or may not belong to that set.

$\Rightarrow$ Least upper bound property;

An order set S is said to have the least-upper bound property if the following is true;

If $E \subset S$, E is not empty, and E is bounded above, then $\sup E$ exists in S .

$\Rightarrow$ Theorem; (The Completeness Axiom)

If a non-empty set of Real Numbers is bounded above, then it has a supremum, this is called completeness.

$\Rightarrow$ Real number system is a complete ordered field.

$\Rightarrow$ Theorem : - If a non-empty set S of Real number is bounded above, then $\sup S$ is the unique real number β , |
• $x \leq \beta$ for all x in S .

* If $\epsilon > 0$, there is an x_0 in S such that $x_0 > \beta - \epsilon$.

⑤ $\Rightarrow$ The Archimedean property of $\mathbb{R}$:

Theorem; If p and ϵ are positive, then
 $n\epsilon > p$ for some integer n , $n\epsilon \leq p$

$\Rightarrow$ Dense set in $\mathbb{R}$:

A set D is said to be dense in the set of Real numbers if every open interval (a, b) contains a number of D .

$\Rightarrow$ Theorem;

The Rational Numbers are dense in $\mathbb{R}$, that is, if a & b are Real numbers with $a < b$, there is a Rational number p/q such that $a < p/q < b$.

$\Rightarrow$ Set of Rational Number is not complete.

Theorem;

product of a Rational and irrational number is irrational.

Proof;

Let r is Rational and r' is irrational number; $r = p/q$
So we can write down as;

$$r r' = \frac{m}{n} \quad \Rightarrow \quad m \neq 0, \quad q \neq 0, \quad r' \neq 0$$

$$r' = \frac{am}{bn} \text{ it's contradiction.}$$

So r' is irrational number:-

Theorem;

Sum of Rational and irrational Number is irrational.

Proof;

let r is Rational and r' is an irrational Number,

$$r + r' = \frac{m}{n}$$

Because r is rational number so,

$$r = \frac{p}{q}$$

$$r' = \frac{m}{n} - \frac{p}{q} \Rightarrow r' = \frac{m-p}{nq}$$

it is contradiction.

So $r + r'$ is an irrational.

Theorem;

For every real $x > 0$ and every integer $n > 0$ there is one and only one positive real y such that $y^n = x$

⑥ Theorem;

if a and b are any two Real Numbers, then

$$|a+b| \leq |a| + |b|$$

$\Rightarrow$ The Extended Real Number System;

* A nonempty set S of Real numbers is unbounded above if it has no upper bound.

* A nonempty set S of Real numbers is unbounded below if it has no lower bound.

* unbounded above;

$$\inf S = -\infty \quad \sup S = \infty$$

* unbounded below;

$$\inf S = -\infty$$

$\Rightarrow$ Mathematical Induction Example:-

Example;

$$1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

$$P_1 \Rightarrow P(1)$$

$$1 = \frac{1(1+1)}{2} \Rightarrow 1 = \frac{1(2)}{2} \Rightarrow 1 = \frac{2}{2} = 1$$

$$P_2 \Rightarrow \text{put } n = n+1$$

$$(1 + 2 + \dots + n) + (n+1) = \frac{(n+1)(n+1+1)}{2} = \frac{(n+1)(n+2)}{2}$$

Example;

For each non-negative integer n , let x_n be a Real number and suppose that;

$$|x_{n+1} - x_n| \leq r |x_n - x_{n-1}|, n \geq 1 \quad \text{--- (1)}$$

Show that;

$$|x_n - x_{n-1}| \leq r^{n-1} |x_1 - x_0| \quad \text{i.f. } n \geq 1 \rightarrow \text{--- (2)}$$

Proof

For $n = 1, 2, 3$ we define

$$|x_{1+1} - x_1| \leq r |x_1 - x_{1-1}| \quad \text{i.f. } n \geq 1$$

$$|x_2 - x_1| \leq r |x_1 - x_0|,$$

$$|x_3 - x_2| \leq r |x_2 - x_1| \leq r^2 |x_1 - x_0|$$

$$|x_4 - x_3| \leq r |x_3 - x_2| \leq r^3 |x_1 - x_0|$$

Inductive step for $P(n+1)$

put $n = n+1$ then we have;

$$|x_{n+1+1} - x_{n+1}| \leq r |x_{n+1} - x_{n+1-1}|$$

$$|x_{n+2} - x_{n+1}| \leq r |x_{n+1} - x_n| \Rightarrow \text{eq. (A)}$$

put in eq. (2) $n = n+1$

$$|x_{n+1} - x_{n+1-1}| \leq r^{n+1-1} |x_1 - x_0|$$

$$|x_{n+1} - x_n| \leq r^n |x_1 - x_0| \rightarrow \text{--- (B)}$$

Comparing (A) & (B)

$$|x_{n+2} - x_{n+1}| \leq |x_{n+1} - x_n| \leq r^n |x_1 - x_0|$$

Hence proved.

⑦

Example:-

prove the proposition;

$$3n + 16 > 0, n \geq -5$$

Solve)

$$n = -5, -4, -3, \dots$$

for $P(-5)$ up to $n = -5$

$$3(-5) + 16 > 0$$

$$-15 + 16 > 0$$

$$1 > 0$$

Basis step is true.

put $n = n+1$

$$3(n+1) + 16 > 0$$

$$3n + 16 + 3 > 0$$

$$3n + 19 > 0$$

Inductive step is also true for given proposition

Example:-

let P_n be the proposition that;

$$n! - 3^n > 0, n \geq 7$$

Sol:- ① P_n is true when we put $n = 7$

② For $P(n+1)$

$$(n+1)! - 3^{n+1} > 0 \Rightarrow n!(n+1) - 3^{n+1} > 0$$

$$3^n(n+1) - 3^{n+1} > 0$$

$$3^n(n+1) - 3^n \cdot 3 > 0$$

$$3^n(n+1-3) > 0 \Rightarrow 3^n(n-2) > 0$$

$$\because n! - 3^n > 0$$

$$n! = 3^n$$

so inductive step is also True.

=> Generalization of union and Intersection;

* The union and Intersection of finitely many sets are written as;

$$\bigcup_{k=1}^n S_k \text{ and } \bigcap_{k=1}^n S_k.$$

* The union and Intersection of an infinite sequence $\{S_k\}_{k=1}^{\infty}$ of sets written as,

$$\bigcup_{k=1}^{\infty} S_k \text{ and } \bigcap_{k=1}^{\infty} S_k.$$

* The open Interval (a, ∞) and $(-\infty, b)$ are semi-infinite.

* If a , and b are finite, and $(-\infty, \infty)$ is then entire real line.

=> ϵ - Neighborhood:-

(If x_0 is a real number and $\epsilon > 0$ then the open Interval $(x_0 - \epsilon, x_0 + \epsilon)$ is an ϵ -Neighborhood of x_0 .)

=> Deleted Neighborhood:-

A Deleted neighborhood of a point x_0 is a set that contains every point of some neighborhood of x_0 except for x_0 itself.

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$\Rightarrow$ The union of open set is open.

$\Rightarrow$ The intersection of closed sets is closed.

$\Rightarrow x_0$ is a limit point of S if every deleted Neighborhood of x_0 contains a point of S .

$\Rightarrow x_0$ is a boundary point of S if every Neighborhood of x_0 contains at least one point in S and one not in S .
Denoted by ∂S .

$\Rightarrow$ The Closure of S , denoted by $\bar{S}$.

$$\bar{S} = S \cup \partial S$$

$\Rightarrow x_0$ is an isolated point of S if $x_0 \in S$ and there is a Neighborhood of x_0 that contains no other point of S .

$\Rightarrow x_0$ is exterior to S if x_0 is in the interior of S^c . The collection of such points is the exterior of S .

$\Rightarrow$ Open Covering:-

A Collection $\mathcal{H}$ of open sets is an open covering of a set S if every point in S is contained in a set H belonging to $\mathcal{H}$,

$$S \subset \bigcup \{H \in \mathcal{H} : H \in \mathcal{H}\}$$

$\Rightarrow$ Heine-Borel theorem;

iff $\mathcal{H}$ is an open covering of a closed and bounded subset S of the Real number then S has an open covering $\mathcal{H}$ consisting of finitely many open sets belonging to $\mathcal{H}$.

$\Rightarrow$ Compact set;

A closed and bounded set of real numbers is called compact.

$\Rightarrow$ Theorem;

A set is closed iff no point of S^c is a limit point of S .

$\Rightarrow$ A set S is closed iff S contains all of its limit points.

$\Rightarrow$ (Bolzano Weierstrass Theorem)

"Every Bounded infinite set of Real numbers has at least one limit point."

⑨ Chapter # 02

Sequences and Series

⇒ Sequences:-

"An infinite sequence of real numbers is a real-valued function defined on a set of integers $n: n \geq k$."

MCQs:

$$\{s_n\}_k^\infty = \{s_k, s_{k+1}, \dots\}$$

$$\{(-1)^n\}_0^\infty = \{1, -1, 1, \dots, (-1)^n, \dots\},$$

$$\left\{\frac{1}{n-2}\right\}_3^\infty = \left\{1, \frac{1}{3}, \frac{1}{3}, \dots, \frac{1}{n-2}, \dots\right\}$$

The real number s_n is the n th term of the sequence.

$$\left\{\frac{1}{n-2}\right\}_3^\infty \text{ and } \left\{\frac{1}{n}\right\}_1^\infty \text{ are identical.}$$

⇒ Convergent Sequence:-

A sequence $\{s_n\}$ converge to a limit L , if for every $\varepsilon > 0$ there is an Integer N such that;

$$|s_n - L| < \varepsilon \text{ if } n \geq N$$

s_n is convergent and write;

$$\lim_{n \rightarrow \infty} s_n = L.$$

* A sequence that does not converge is called diverges or is divergent.

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$\Rightarrow$ The limit of a convergent sequence is unique.

$\Rightarrow$ A sequence $\{s_n\}$ is said to be divergent to ∞ if for any Real number a , $s_n > a$ for large n and written as;

$$\lim_{n \rightarrow \infty} s_n = \infty$$

Similarly $\Rightarrow \lim_{n \rightarrow \infty} s_n = -\infty$ if for any Real number a , $s_n < a$ for large n .

Example:-

The sequence $\left\{ \frac{n}{2} + \frac{1}{n} \right\}$ diverges to ∞ , since, if a is any Real number, then

$$\frac{n}{2} + \frac{1}{2} > a \text{ if } n \geq 2a$$

$$\lim_{n \rightarrow \infty} \left(\frac{n}{2} + \frac{1}{n} \right) = \infty$$

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Bounded Sequence,

A sequence (s_n) is bounded above if there is a Real Number b such that

$$s_n \leq b \text{ for all } n,$$

Bounded below if there is a real Number a such that;

$$s_n \geq a \text{ for all } n.$$

$\Rightarrow$ A Convergent sequence is bounded.

$\Rightarrow$ A sequence $\{s_n\}$ converges to s if and only if every neighborhood of s contains s_n for all but finitely many n .

$\Rightarrow$ Monotonic sequences;

"A sequence that is either nonincreasing or Nondecreasing is called Monotonic sequences."

* A sequence $\{s_n\}$ is nondecreasing if $s_n \geq s_{n-1}$ for all n .

* A sequence $\{s_n\}$ is non-Increasing if $s_n \leq s_{n-1}$ for all n .

Theorem:-

* If $\{s_n\}$ is non-decreasing then
 $\lim_{n \rightarrow \infty} s_n = \sup \{s_n\}$

* If $\{s_n\}$ is non-increasing, then $\lim_{n \rightarrow \infty} s_n = \inf \{s_n\}$.

$\Rightarrow$ If $s_n > s_{n-1}$ then $\{s_n\}$ is increasing.

$\Rightarrow$ If $s_n < s_{n-1}$ then $\{s_n\}$ is decreasing.

Theorem;

If $p > 0$ then

$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$$

Inequality From Binomial theorem;

$$1 + nx \leq (1+x)^n, \quad x > 0$$

Theorem; show that;

$$\lim_{n \rightarrow \infty} n\sqrt[n]{n} = 1$$

Proof;

$$\text{Take } x_n = n\sqrt[n]{n} - 1$$

Then $x_n \geq 0$ and by the binomial theorem;

$$(1+x_n)^n \geq \frac{n(n-1)}{2} x_n^2$$

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$$x_n^2 \leq \frac{2}{n-1}$$

$$0 \leq x_n \leq \sqrt{\frac{2}{n-1}}, \quad n \geq 2$$

i.e., we take limit;

$$\lim_{n \rightarrow \infty} \sqrt{\frac{2}{n-1}} = 0$$

$$\text{So, } 0 \leq x_n \leq 0$$

$$\text{So } \lim_{n \rightarrow \infty} x_n = 0.$$

Theorem;

i.e., $p > 0$ and α is real, the

$$\lim_{n \rightarrow \infty} \frac{n^\alpha}{(1+p)^n} = 0$$

Proof;

let k be an integer such that

$$k > \alpha, \quad k > 0$$

$$\because n(n-1) \cdots n-(k+1) \geq \frac{n!}{2^k}$$

for $n > 2k$.

$$(1+p)^n > \frac{n(n-1) \cdots n-(k+1)}{k!} p^k > \frac{n! p^k}{2^k k!}$$

$$(1+p)^n > \frac{n! p^k}{2^k k!} \Rightarrow \text{By using Reciprocal;}$$

$$\frac{1}{(1+b)^n} > \frac{2^{k!}}{n^k p^k}$$

$$\frac{1}{(1+b)^n} < \frac{n^{-k} 2^{k!}}{p^k}$$

$$\frac{1 \times n^\alpha}{(1+b)^n} < \frac{n^{\alpha-k} 2^{k!}}{p^k}$$

Hence;

$$0 < \frac{n^\alpha}{(1+b)^n} < \frac{2^{k!}}{p^k} n^{\alpha-k}, \quad n > 2k$$

Since $\alpha - k < 0$, $n^{\alpha-k} \rightarrow 0$ by

$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$$

Hence proved;

$$\lim_{n \rightarrow \infty} \frac{n^\alpha}{(1+b)^n} = 0$$

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Theorem;

Let $\lim_{n \rightarrow \infty} S_n = s$ and $\lim_{n \rightarrow \infty} t_n = t$ where

s and t are finite.

then

$$\lim_{n \rightarrow \infty} (cS)_n = cS$$

if c is constant;

$$\lim_{n \rightarrow \infty} (S_n + t_n) = s + t$$

$$\lim_{n \rightarrow \infty} (S_n - t_n) = s - t$$

Example:-

Find the limit of the sequence

$$S_n = \frac{1}{n} \frac{\sin n\pi}{4} + \frac{2(1+3)/n}{1+1/n}$$

Solve;

we apply the applicable parts of

Theorem;

$$\begin{aligned} \lim_{n \rightarrow \infty} S_n &= \lim_{n \rightarrow \infty} \frac{1}{n} \frac{\sin n\pi}{4} + \frac{2\left(\lim_{n \rightarrow \infty} 1 + 3 \lim_{n \rightarrow \infty} (1/n)\right)}{\lim_{n \rightarrow \infty} 1 + \lim_{n \rightarrow \infty} (1/n)} \\ &= 0 + \frac{2(1+3 \cdot 0)}{1+0} = 2 \end{aligned}$$

Example:-

Find the limit of the Subsequence

$$S_n = \lim_{n \rightarrow \infty} \frac{(n/2) + (\log n)}{3n + 4\sqrt{n}}$$

$$= \lim_{n \rightarrow \infty} \frac{1/2 + (\log n)/n}{3 + 4n^{-1/2}}$$

$$= \frac{\lim_{n \rightarrow \infty} 1/2 + \lim_{n \rightarrow \infty} (\log n)/n}{\lim_{n \rightarrow \infty} 3 + 4 \lim_{n \rightarrow \infty} n^{-1/2}}$$

$$= \frac{1/2 + 0}{3 + 0}$$

$$= 1/6$$

Subsequence:-

A sequence $\{t_k\}$ is a subsequence of a sequence $\{s_n\}$ if;
 $t_k = s_{n_k} \quad k \geq 0$

where $\{n_k\}$ is an increasing infinite sequence of integers in the domain of $\{s_n\}$.

we denote the subsequence $\{t_k\}$ by $\{s_{n_k}\}$

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Example:

$$\text{ID } \{s_n\} = \left\{ \frac{1}{n} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, \dots \right\}$$

let,

$$n_k = 2k$$

$$\{s_{2k}\} = \left\{ \frac{1}{2k} \right\} = \left\{ \frac{1}{2}, \frac{1}{4}, \dots, \frac{1}{2k}, \dots \right\}$$

$$n_k = 2k+1$$

$$\{s_{2k+1}\} = \left\{ \frac{1}{2k+1} \right\} = \left\{ 1, \frac{1}{3}, \frac{1}{5}, \dots, \frac{1}{2k+1}, \dots \right\}$$

Theorem:

"Every subsequence of convergent sequence is convergent."

Theorem:

"If a sequence is Monotonic & has convergent subsequence then the sequence is convergent."

$$\lim_{k \rightarrow \infty} s_{n_k} = s,$$

then $\lim_{n \rightarrow \infty} s_n = s.$

Limit point;

Theorem;

A point $\bar{x}$ is a limit point of a set S iff there is a sequence $\{x_n\}$ of point in S such that $x_n \neq \bar{x}$ for $n \geq 1$ and

$$\lim_{n \rightarrow \infty} x_n = \bar{x}.$$

Theorem:-

* If $\{x_n\}$ is bounded, then $\{x_n\}$ has a convergent subsequence.

* If $\{x_n\}$ is unbounded above, then $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ such that ;

$$\lim_{k \rightarrow \infty} x_{n_k} = \infty$$

* If $\{x_n\}$ is unbounded below, then

$\{x_n\}$ has a subsequence $\{x_{n_k}\}$ such that ;

$$\lim_{k \rightarrow \infty} x_{n_k} = -\infty$$

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limit Superior and limit Inferior;Theorem:-

* If $\{s_n\}$ is Bounded above and does not diverge to $-\infty$, then there is a unique Real Number $\bar{s}$, such that, if $\varepsilon > 0$

$$s_n < \bar{s} + \varepsilon \quad \text{for large } n$$

and

$$s_n > \bar{s} - \varepsilon \quad \text{for infinitely many } n.$$

* If $\{s_n\}$ is bounded below and does not diverge to ∞ , then there is a unique Real Number $\underline{s}$ such that, if $\varepsilon > 0$,

$$s_n > \underline{s} - \varepsilon \quad \text{for large } n$$

$$s_n \leq \underline{s} + \varepsilon \quad \text{for infinitely many } n.$$

limit superior and limit inferior of $\{s_n\}$ is denoted by;

$$\bar{s} = \limsup_{n \rightarrow \infty} s_n$$

$$\underline{s} = \liminf_{n \rightarrow \infty} s_n$$

$\Rightarrow \limsup_{n \rightarrow \infty} s_n = \infty \Rightarrow$ if $\{s_n\}$ is not bounded above

$\Rightarrow \limsup_{n \rightarrow \infty} s_n = -\infty \Rightarrow$ if $\lim_{n \rightarrow \infty} s_n = -\infty$

$\Rightarrow \liminf_{n \rightarrow \infty} s_n = -\infty \Rightarrow$ if $\{s_n\}$ is not bounded ^{below} above

$\Rightarrow \liminf_{n \rightarrow \infty} s_n = \infty \Rightarrow$ if $\lim_{n \rightarrow \infty} s_n = \infty$

Theorem;

Every sequence $\{s_n\}$ of Real Numbers has a unique limit Superior, $\bar{s}$ and limit inferior, $\underline{s}$, in the extended Reals,

$$\underline{s} \leq \bar{s}.$$

Cauchy Sequence;

A sequence $\{s_n\}$ of Real numbers is called a Cauchy sequence if for every $\epsilon > 0$, there is an integer N that, is

$$|s_n - s_m| < \epsilon \text{ if } m, n \geq N.$$

Theorem;

if $\{s_n\}$ is a Cauchy sequence of Real Numbers, then $\{s_n\}$ is bounded.

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Theorem;

A sequence $\{s_n\}$ of Real Numbers Converges $\iff$ for every $\epsilon > 0$ there is an integer N such that;

$$|s_n - s_m| < \epsilon \quad \text{if } m, n \geq N$$

Series

Topic #60

 $\Rightarrow$ **oscillatory series;**

A divergent infinite series that does not diverge to $\pm\infty$ is said to oscillate, or the oscillatory.

 $\Rightarrow$ **Theorem;**

The sum of a convergent series is unique.

$$\Rightarrow \sum_{n=k}^{\infty} (c a_n) = cA$$

if c is constant,

$$\Rightarrow \sum_{n=k}^{\infty} (a_n + b_n) = A + B$$

$$\Rightarrow \sum_{n=k}^{\infty} (a_n - b_n) = A - B$$

$\Rightarrow$ The Series $\sum_k a_n$ Converges iff $(-1)^n a_n > 0$,
 $|a_{n+1}| < |a_n|$, and $\lim_{n \rightarrow \infty} a_n = 0$

$\Rightarrow$ Theorem;

A Series $\sum a_n$ Converges iff for every $\varepsilon > 0$ there is an Integer N such that;

$$|a_n + a_{n+1} + \dots + a_m| < \varepsilon \text{ if } m \geq n \geq N.$$

$\Rightarrow \sum a_n$ Converges iff $\{A_n\}$ Converges.

Theorem;

Show that the Series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent

$$S_1 = 1, S_2 = 1 + \frac{1}{2}, S_3 = 1 + \frac{1}{2} + \frac{1}{3}, S_4 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}, \dots$$

Form strictly increasing sequence.

$$S_1 < S_2 < S_3 < S_4 < \dots < S_n < \dots$$

$$S_2 - S_1 = 1 + \frac{1}{2} - 1 = \frac{1}{2}$$

$$S_4 - S_2 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} - (1 + \frac{1}{2}) = \frac{1}{3} + \frac{1}{4} > S_2 - S_1 = \frac{1}{2}$$

$$(16) \quad S_8 = 2^3 = S_4 + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} > S_4 + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} > \frac{4}{8}$$

$$\sim S_{2n} > \frac{n+1}{2}$$

If M is any constant, we can find a positive integer n such that;

$$\frac{n+1}{2} > M.$$

for n we have

$$S_{2n} > \frac{n+1}{2} > M$$

Example,

If $\sum a_n$ Converges, then $\lim_{n \rightarrow \infty} a_n = 0$

Proof,

If $\sum a_n$ Converges, then for each $\epsilon > 0$ there is an integer K such that;

$$\left| \sum_{n=K}^{\infty} a_n \right| < \epsilon \quad \text{if } K \geq K_0;$$

that is

$$\lim_{K \rightarrow \infty} \sum_{n=K}^{\infty} a_n = 0$$

$\lim_{n \rightarrow \infty} a_n = 0$ Necessary Condition not Sufficient.

Series of Non-Negative;

"The Series $\sum a_n$ is said to be Series of Non-Negative terms if $a_n \geq 0$ for $n \geq k$."

"The Series of Non-Negative term either converges to a finite limit or diverges to ∞ ."

The Comparison Test

Theorem

Suppose that

$$0 \leq a_n \leq b_n, \quad n \geq k,$$

Then;

$$\star \quad \sum a_n < \infty \text{ if } \sum b_n < \infty$$

$$\star \quad \sum b_n = \infty \text{ if } \sum a_n = \infty$$

Proof :-

if;

$$A_n = a_k + a_{k+1} + \dots + a_n \text{ and}$$

$$B_n = b_k + b_{k+1} + \dots + b_n, \quad \underline{n \geq k}$$

then, we have;

$$A_n \leq B_n$$

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$\Rightarrow$ If $\sum b_n < \infty$, then $\{B_n\}$ is Bounded above
 this implies that $\{A_n\}$ is also, therefore
 $\sum a_n < \infty$.

$\Rightarrow$ If $\sum a_n = \infty$ then $\{A_n\}$ is unbounded above
 implies that $\{B_n\}$ is also, therefore, $\sum b_n = \infty$

Example:-

Discuss the Convergence of the
 Series $\sum \frac{x^n}{n}$.

Since,

$$\frac{x^n}{n} < x^n, \quad n \geq 1$$

$\sum x^n < \infty$ ~~if~~ if $0 < x < 1$, the Series
 $\sum x^n/n$ converges if $0 < x < 1$.

The Series is inconclusive (Mean we can't
 say anything about Convergent & divergent
 of Series) if $x > 1$

If $x < 0$ Comparison test does not apply.
 Since the Series have infinitely many
 Negative terms.

Theorem;

Suppose $a_1 \geq a_2 \geq \dots \geq 0$. Then the Series $\sum_{n=1}^{\infty} a_n$ Converges iff the Series $\sum_{k=0}^{\infty} 2^k a_{2^k}$ Converges.

Theorem;

The Series $\sum \frac{1}{n^p}$ Converges if $p > 1$ and diverges if $p \leq 1$

Theorem;

if $p > 1$
$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$$
 Converges.
if $p < 1$, the Series diverges.

Proof:-

The Monotonicity of the logarithmic Function implies $\{\log n\}$ increases. Hence $\{1/n \log n\}$ decreases.

$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$$

By using theorem;

⑩ ∴ Suppose $a_1 \geq a_2 \geq \dots \geq 0$ then the Series $\sum_{n=1}^{\infty} a_n$ converges iff the Series $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.

$$a_n = \frac{1}{n(\log n)^p} \Rightarrow a_{2^k} = \frac{1}{2^k (\log 2^k)^p}$$

$$\Downarrow$$

∴ By using log property;
 $\Rightarrow \log m^n = n \log m$

So the Series;

$$a_{2^k} = \frac{1}{2^k (k \log 2)^p}$$

$$\sum_{k=1}^{\infty} \frac{2^k}{2^k (k \log 2)^p} = \frac{1}{(\log 2)^p} \sum_{k=1}^{\infty} \frac{1}{k^p}$$

So By using the result of Harmonic Series.

our given Series is Convergent
 iff $p > 1$ and divergent $p < 1$.

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(The Number e)

Imp long

The number e is defined by infinite series and is given by;

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}$$

where; $n! = 1 \cdot 2 \cdot 3 \cdots n$ if $n \geq 1$ and $0! = 1$

(Approximation of Number 'e')

Imp long

Let the sequence of partial sum!

$$S_n = 1 + 1 + \frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 2 \cdot 3} + \cdots + \frac{1}{1 \cdot 2 \cdot 3 \cdots n}$$

if we subtract (e) from this partial sum

All term upto (n) terms will be cancel out and then we have;

$$e - S_n = \frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \frac{1}{(n+3)!} + \cdots$$

$$< \frac{1}{(n+1)!} \left(1 + \frac{1}{n+1} + \frac{1}{(n+1)^2} + \cdots \right) = \frac{1}{n! n}$$

so, we can conclude that;

$$0 < e - S_n < \frac{1}{n! n}$$

$\Rightarrow$ The number e is an irrational number.

Theorem;

Suppose that $a_n \geq 0$ and $b_n > 0$ for $n \geq k$. Then;

* $\sum a_n < \infty$ iff $\sum b_n < \infty$ and $\limsup_{n \rightarrow \infty} a_n/b_n < \infty$

* $\sum a_n = \infty$ iff $\sum b_n = \infty$ and $\liminf_{n \rightarrow \infty} a_n/b_n > 0$

Example;

$$\sum b_n = \sum \frac{1}{n^{p+q}} \quad \text{and} \quad \sum a_n = \sum \frac{2 + \sin n\pi/6}{(n+1)^p (n-1)^q}$$

Sol;

$$\frac{a_n}{b_n} = \frac{2 + \sin n\pi/6}{(1 + 1/n)^p (1 - 1/n)^q}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 3, \quad \lim_{n \rightarrow \infty} \frac{a_n}{b_n} \geq 1.$$

Since $\sum b_n < \infty$ iff $p+q > 1$, the same is true of $\sum a_n$.

20/ Topic 80

⇒ The Ratio Test:-

Theorem: Suppose that $a_n > 0$, $b_n > 0$ and

$$\frac{a_{n+1}}{a_n} \leq \frac{b_{n+1}}{b_n}$$

we can also write it as;

$$\frac{a_{n+1}}{b_{n+1}} \leq \frac{a_n}{b_n} \Rightarrow \left\{ \frac{a_n}{b_n} \right\} \text{ is Non-Increasing.}$$

(Topic 81) Then:

$$\ast \sum a_n < \infty \text{ if } \sum b_n < \infty$$

$$\ast \sum b_n = \infty \text{ if } \sum a_n = \infty$$

⇒ Suppose that $a_n > 0$ ($n \geq k$) and

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = L$$

Then

$$\sum a_n < \infty \text{ if } L < 1.$$

$$\sum a_n = \infty \text{ if } L > 1.$$